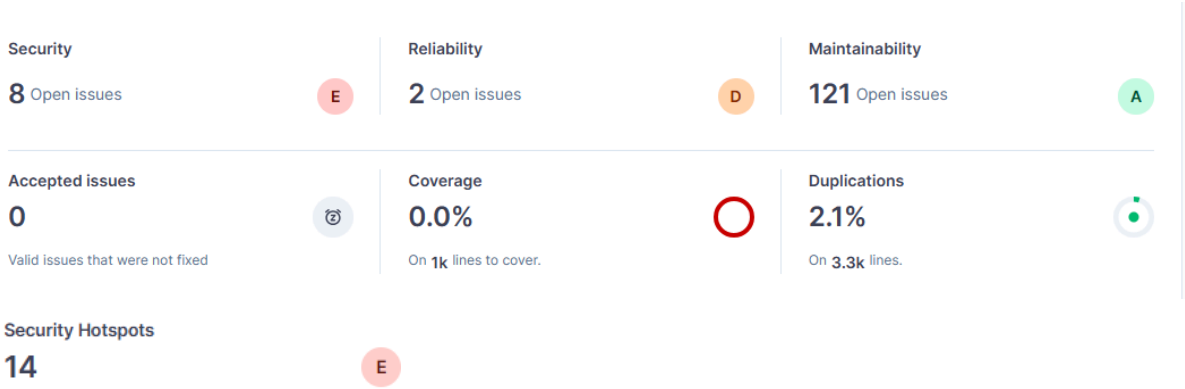
Initial State:

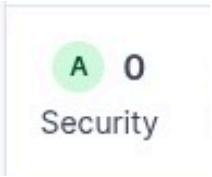


Security issues:

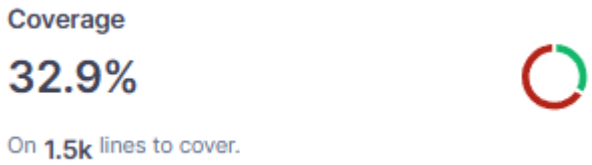
The security issues are due to a hard-coded password in database initiation.

The security hotspots concern the use of a pseudorandom number generator in the game logic and the printStackTrace() debug feature in the database handler.

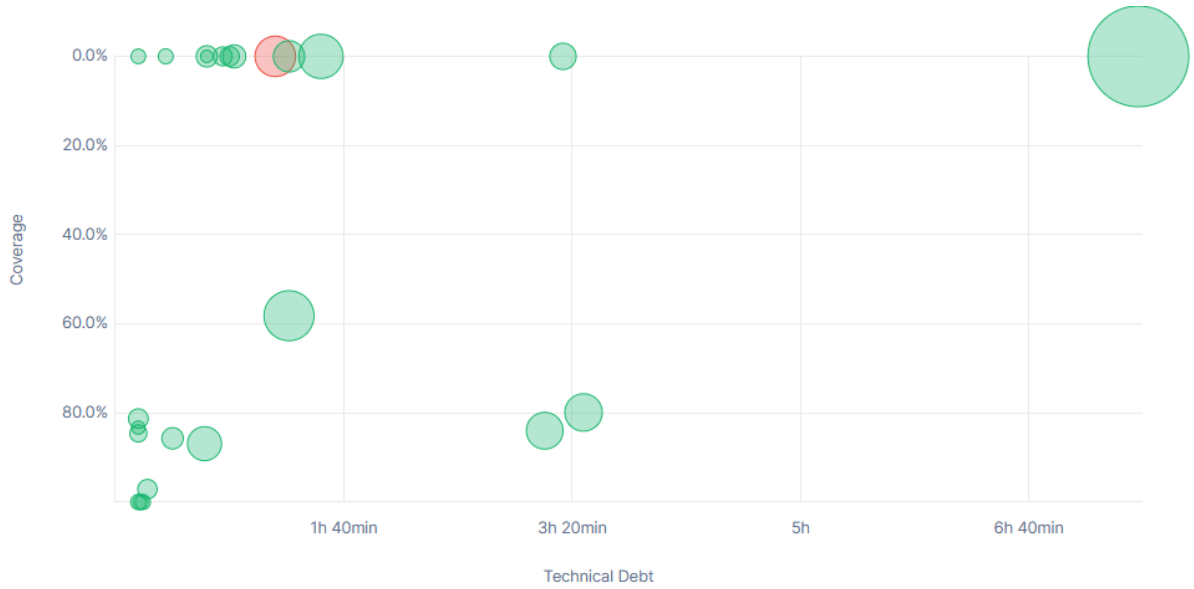
After fixes:



Test Coverage:



Initial total test coverage of 32.9%



The area with a large technical Debt and no test coverage is the main game UI class, while the unsecure area (in red) is the database handler.
Test coverage after code clean-up

Remaining uncovered code consists of methods that utilize javaFX or the database. Testing them requires specialized proxies for javaFX and the database.

Reliability issues:

Making a single random generator

carcassonne/.../main/java/carcassonne/Model/Game.java

Save and re-use this "Random".

Reliability High

```
254        Random rand = new Random();
   juan.v…
255        int size = deck.size();
```

Save and re-use this "Random".

Overriding hashCode()

carcassonne/.../main/java/carcassonne/Model/Spot.java

This class overrides "equals()" and should therefore also override "hashCode()".

Reliability Low

```
50         @Override
51         public boolean equals(Object obj) {
   juan.v…
```

This class overrides "equals()" and should therefore also override "hashCode()".

After fix:

Reliability

0 Open issues A